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| HALF YEARLY EXAMINATION - 2025 | 12 - STD | |
|---|---|---|
| MATHEMATICS | Marks 90 | Time 3.00 Hrs. |

PART - A

I. Choose the correct answer. 20 x 1 = 20

1. If $A=\begin{bmatrix}2 & 0\\ 1 & 5\end{bmatrix}$, $B=\begin{bmatrix}1 & 4\\ 2 & 0\end{bmatrix}$, then $|adj\,(AB)|$ = .....
   a) -40 b) -80 c) -60 d) -20
2. If Z is a non-zero complex number, such that $2iZ^2=\overline{Z}$ then $|Z|$ is .....
   a) $\frac{1}{2}$ b) 1 c) 2 d) 3
3. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is .....
   a) 2 b) 4 c) 1 d) $\infty$
4. $\sin(\tan^{-1} x)$, $|x| < 1$ is equal to
   a) $\frac{x}{\sqrt{1-x^2}}$ b) $\frac{1}{\sqrt{1-x^2}}$ c) $\frac{1}{\sqrt{1+x^2}}$ d) $\frac{x}{\sqrt{1+x^2}}$
5. The accentricity of the ellipse $(x-3)^2+(y-4)^2=\frac{y^2}{9}$ is
   a) $\frac{1}{3}$ b) $\frac{\sqrt{3}}{2}$ c) $\frac{1}{\sqrt{3}}$ d) $\frac{1}{3\sqrt{2}}$
6. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}+\hat{j}$, $\vec{c}=\hat{i}$ and $(\vec{a}\times\vec{b})\times\vec{c}=\lambda\vec{a}+\mu\vec{b}$, then the value of $\lambda+\mu$ is .....
   a) 0 b) 6 c) 1 d) 3
7. Angle between $y^2=x$ and $x^2=y$ at the origin is .....
   a) $\tan^{-1}\left(\frac{4}{3}\right)$ b) $\frac{\pi}{2}$ c) $\tan^{-1}\left(\frac{3}{4}\right)$ d) $\frac{\pi}{4}$
8. If $f(x, y, z) = xy + yz + zx$, then $f_x - f_z$ is equal to
   a) $y - x$ b) $y - z$ c) $x - z$ d) $z - x$
9. The value of $\int_0^1 x(1-x)^{99}dx$ is .....
   a) $\frac{1}{10100}$ b) $\frac{1}{10010}$ c) $\frac{1}{11000}$ d) $\frac{1}{10001}$
10. The general solution of the differential equation $\frac{dy}{dx}=\frac{y}{x}$ is
    a) $y = kx$ b) $xy = k$ c) $\log y = kx$ d) $y = k \log x$
11. If $P(x = 0) = 1 - P(x = 1)$. If $E(x) = 3$ var $(x)$, then $P(x = 0)$ is
    a) $\frac{2}{3}$ b) $\frac{2}{5}$ c) $\frac{1}{5}$ d) $\frac{1}{3}$
12. If a compound statement involves 3 simple statements then the number of rows in the truth table is
    a) 8 b) 6 c) 3 d) 9



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13. The number of arbitrary constants in the general solutions of order $n$ and $n+1$ are
a) $n-1, n$ b) $n, n+1$ c) $n+1, n+2$ d) $n+1, n$

14. The approximate change in the value $v$ of a cube of side $x$ metres caused by increasing the side by 1% is
a) $0.03\,x^2 m^3$ b) $0.03\,x^3 m^3$ c) $0.3\,x\,dx\,m^3$ d) $0.03\,x\,m^3$

15. If $cot^{-1} x = \frac{2\pi}{5}$ for some $x \in R$, the value of $tan^{-1} x$ is
a) $\frac{\pi}{5}$ b) $\frac{-\pi}{10}$ c) $\frac{-\pi}{5}$ d) $\frac{\pi}{10}$

16. The value of arg(0) is
a) $\infty$ b) 0 c) $\pi$ d) undefined

17. The length of the latus rectum of the parabola $y^2 - 4x + 4y + 8 = 0$ is
a) -8 b) 6 c) 4 d) 2

18. Find two positive numbers whose sum is 12 and their product is maximum
a) 8, 4 b) 6, 6 c) 4, 8 d) 10, 2

19. Subtraction is not a binary operation in
a) R b) Z c) N d) Q

20. The value of $\int_{-1}^{2} |x|\,dx$ is a) $\frac{1}{2}$ b) $\frac{3}{2}$ c) $\frac{5}{2}$ d) $\frac{7}{2}$

## PART - B

**II. Answer any 7 questions. Q.No. 30 is compulsory.** **7 x 2 = 14**

21. Find the inverse of $\begin{bmatrix} -2 & 4 \\ 1 & -3 \end{bmatrix}$

22. Find $\left|\frac{2+i}{-1+2i}\right|$

23. Show that the equation $2x^2 - 6x + 7 = 0$ cannot be satisfied by any real values of $x$.

24. Find the principal value of $sin^{-1}\left(\frac{-1}{2}\right)$ (in radians and degrees)

25. Identify the type of conic section i) $x^2 + y^2 + x - y = 0$ ii) $y^2 + 4x + 3y + 4 = 0$

26. Prove that $[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}] = 0$

27. Prove that $f(x) = x^2 + 2$ is strictly increasing in (2, 7) and strictly decreasing in (-2, 0).

28. If $W(x, y, z) = x^2 y + y^2 z + z^2 x$, $x, y, z \in R$, find the differential dw.

29. Evaluate : $\int_{-\pi/2}^{\pi/2} x \cos x\,dx$

30. Establish the equivalence property : $p \rightarrow q \equiv \neg p \vee q$

## PART - C

**III. Answer any 7 questions. Q.No.40 is compulsory.** **7 x 3 = 21**

31. Show that $|3z - 5 + i|$ represent a equation of circle and also find its centre and radius.

32. The parabolic communication antenna has a focus at 2 m distance from the vertex of the antenna. Find the width of the antenna 3 m from the vertex.

33. Find the asymptotes of the curve $f(x) = \frac{2x^2 - 8}{x^2 - 16}$



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34. Solve : $\frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$

35. Evaluate : $\int_0^1 x^3(1-x)^4\,dx$

36. A rahdom variable $x$ has the following probability mass function.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $f(x)$ | K | 2K | 6K | 5K | 6K | 10K |

i) Find $K$ ii) Find $P(2<x<6)$

37. Let $A=\begin{bmatrix}0 & 1\\ 1 & 1\end{bmatrix}$ $B=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ be any two boolean matrices of the same type. Find A∧B and A∨B.

38. If $v(x, y, z)=x^3+y^3+z^3+3xyz$ then prove that $\frac{\partial^2 v}{\partial y\partial z}=\frac{\partial^2 v}{\partial z\partial y}$.

39. Find the rank of matrix using Row-Echelon method. $\begin{bmatrix}1 & 2 & 3\\ 2 & 1 & 4\\ 3 & 0 & 5\end{bmatrix}$

40. Prove that $2\tan^{-1}x=\cos^{-1}\left[\frac{1-x^2}{1+x^2}\right]$

## PART - D

**IV. Answer the following.** 7 x 5 = 35

41. a) Test for consistency of the following system of linear equations and solve $x-y+z=-9,\ 2x-2y+2z=-18,\ 3x-3y+3z+27=0$.

(OR)

b) Show that the two curves $x^2-y^2=r^2$ and $xy=c^2$ where $c$, $r$ are constants, cut orthogonally.

42. a) Solve the equation $x^3+27=0$.

(OR)

b) If $u=\sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, then prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial x}=\frac{1}{2}\tan u$

43. a) Solve : $6x^4-35x^3+62x^2-35x+6=0$

(OR)

b) Evaluate : $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$

44. a) Find : $\sin\left[\sin^{-1}\left(\frac{3}{5}\right)+\sec^{-1}\left(\frac{5}{4}\right)\right]$

(OR)

b) The rate of increase in the number of bacteria in a certain bacteria culture is proportional to the number present. Given that the number triples in 5 hours, find how many bacteria will be present after 10 hours.



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45. a) Identify the type of conic and find centre, foci, vertices and directrices of

$$\frac{(x+3)^2}{225}-\frac{(y-4)^2}{64}=1$$

(OR)

b) Find the binomial distribution for each
i) Five fair coins are tossed once and $x$ denotes the number of heads.
ii) A fair die is rolled 10 times and $x$ denotes the number of times 4 appeared.

46. a) Find the non-parametric form of vector equation and cartesian equations of the plane

$$\vec{r}=(6\hat{i}-\hat{j}+\hat{k})+s(-\hat{i}+2\hat{j}+\hat{k})+t(-5\hat{i}-4\hat{j}-5\hat{k})$$

(OR)

b) Verify i) closure property ii) commutative property iii) associative property iv) existence of identify and v) existence of inverse for the operation $+_5$ on $Z_5$ using table corresponding to addition modulo 5.

47. a) Find the area bounded by the curve $y=x^3$ and the line $y=x$.

(OR)

b) If the imaginary part of $\frac{2z+1}{iz+1}$ is -2, then prove that the locus of the point representing $z$ in the complex plan is a straight line.



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